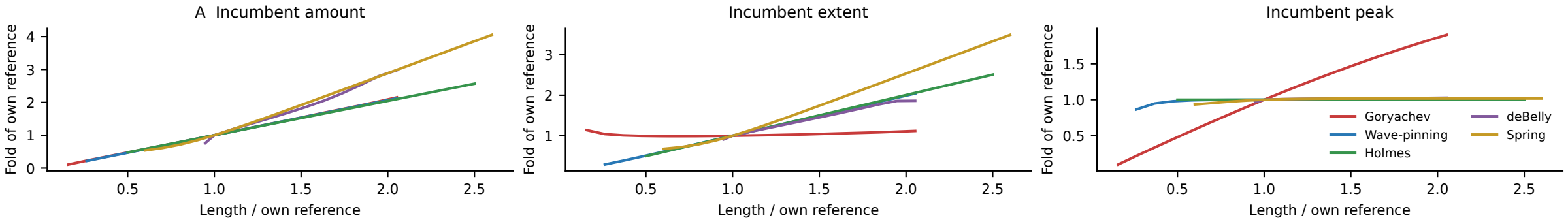
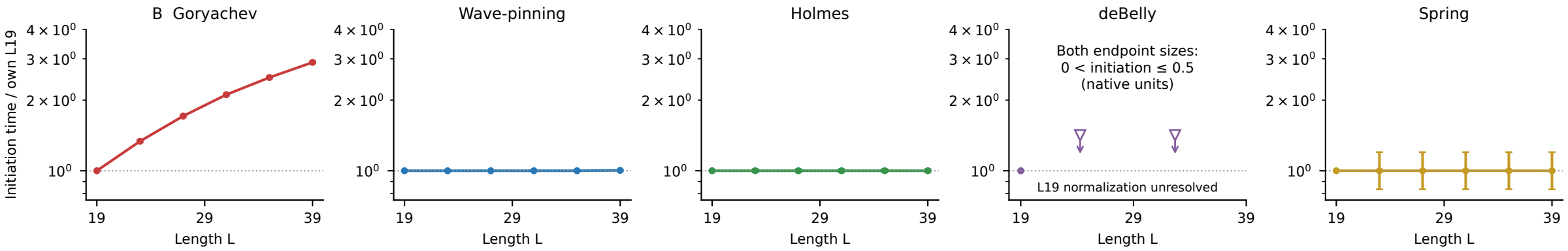
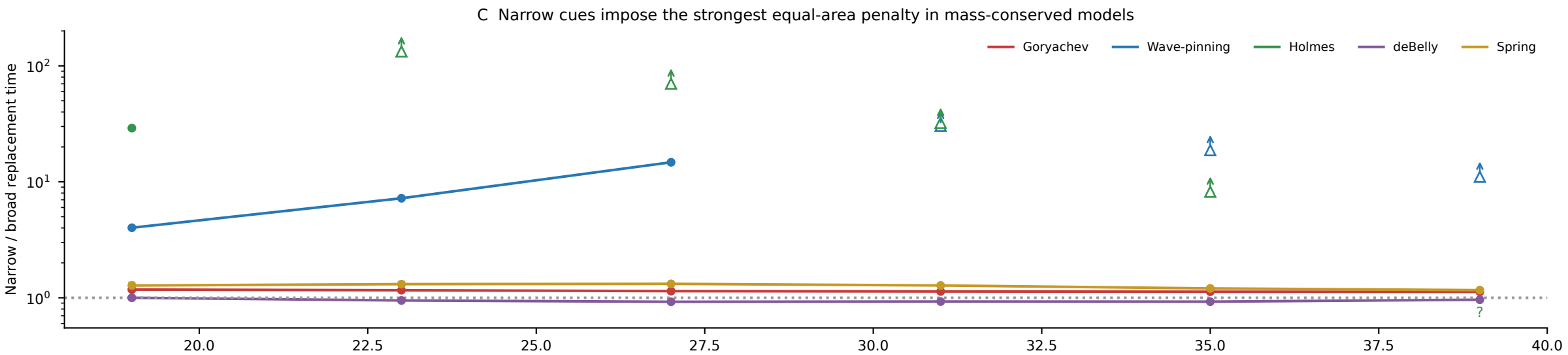
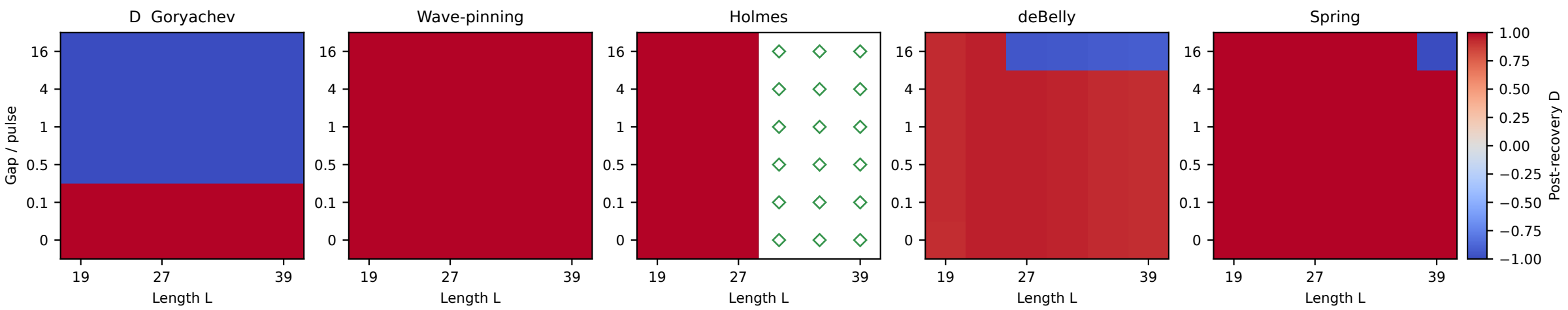
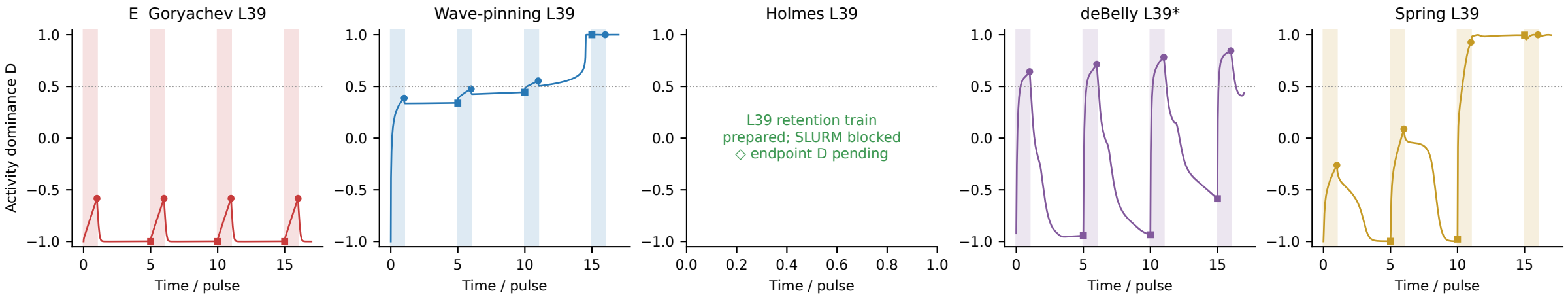
Replacement resistance in enlarged cells arises through three mechanism classes; cue shape and timing distinguish their signatures











F Shape and timing signatures distinguish proposed mechanism classes

Class / models	Initiation slows	Equal-area penalty	Share through gaps	Commit after collapse	Measurement
Initiation-limited Goryachev	Yes	Modest (~1.1×)	Resets	No	Beam width at fixed dose; pulse trains; gap-end share
Mass-conserved, saturating Wave-pinning, Holmes	No	Yes; Holmes growth unresolved at L39	Ratchet; Holmes large L pending	—	Beam width at fixed dose; pulse trains; gap-end share
Slow non-activity state deBelly*, Spring	No resolved slowing*	Modest (~1×)	Collapses early	Yes	Beam width at fixed dose; pulse trains; gap-end share

◇ Holmes large-L retention is unmeasured; ? width ratio uninformative. Open triangles + arrows denote bounds.

* deBelly initiation is interval-censored; later-gap timing and endpoint fields carry numerical qualifications (complete supplement).